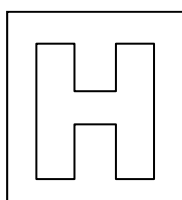


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## 2017 Preliminary Examination 2 Pre-University 3

**H2 Biology**

**9648/03**

**Applications Paper and Planning Question**

**19 September 2017**

**2 hours**

Additional Materials:    Writing paper

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### **READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write your Admission number and name on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a HB pencil for any diagrams or graphs.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.  
You will lose marks if you do not show your working or if you do not use appropriate units.  
At the end of the examination, fasten all your work securely together.  
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |  |
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| 4                  |  |
| 5                  |  |
| Total              |  |

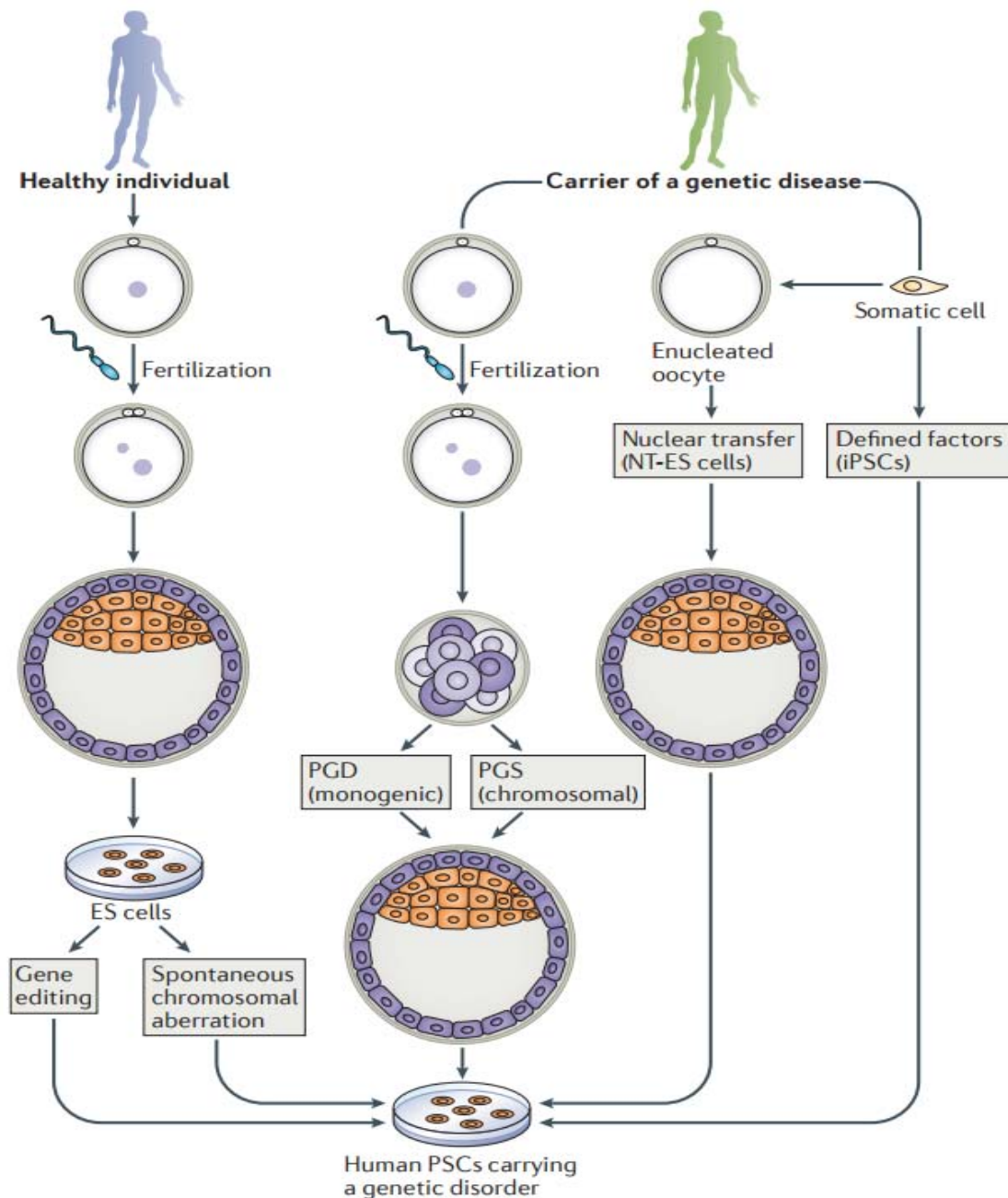
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**This question paper consists of 20 printed pages**

**[Turn over**

Answer **all** questions.

- The ability to model human diseases using cultured pluripotent stem cells (PSCs) has revolutionized the ways in which we study monogenic, complex and epigenetic disorders, as well as early- and late-onset diseases. Several strategies are used to generate such disease models using either embryonic stem cells (ES cells) or patient-specific induced PSCs (iPSCs), creating new possibilities for the establishment of models and their use in drug screening. Fig. 1.1 shows strategies for generating human pluripotent stem cells (hPSCs) carrying a genetic disorder for research purposes. Disease-specific ES cells can be identified during the in-vitro fertilization (IVF) process by pre-implantation genetic diagnosis (PGD) or pre-implantation genetic screening (PGS).



**Fig. 1.1**

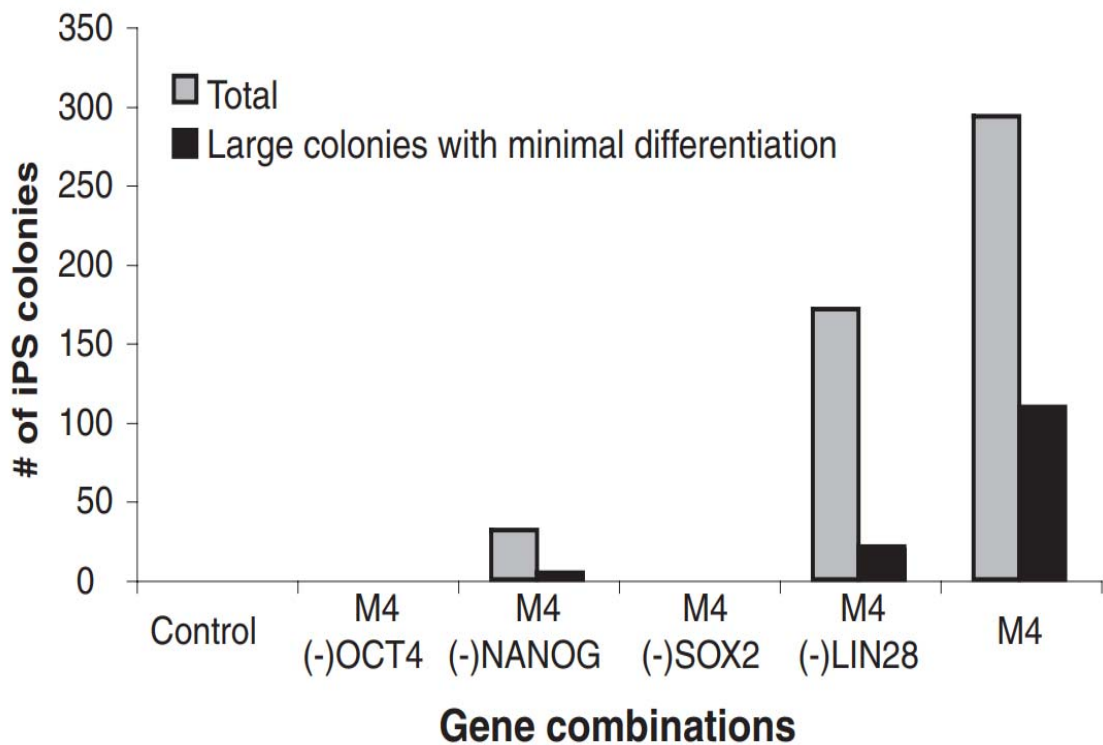
- (a) Describe one similarity and one difference between a blastocyst and embryo.

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 .....  
 .....[2]

- (b) Describe one advantage and one limitation of using somatic cell nuclear transfer (SCNT) to generate human PSCs carrying a genetic disorder.

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 .....[2]

In order to optimise the conditions and increase the chances of creating iPSCs from somatic cells, extensive research has been conducted on 4 main genes, *Oct4*, *Sox2*, *Nanog* and *Lin28* using M4 cell cultures. Fig. 1.2 shows the effect of different combinations of genes in the reprogramming mixture on the number of induced-pluripotent stem cell colonies formed.



**Fig. 1.2**

(c) With reference to Fig. 1.2,

(i) explain the purpose of the control.

.....  
.....[1]

(ii) describe the results produced from varying gene combinations in the reprogramming mixture.

.....  
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.....[2]

(d) Describe one possible regulatory process at the chromosomal level that could increase the expression of *Oct4*, *Sox2*, *Nanog* and *Lin28* genes.

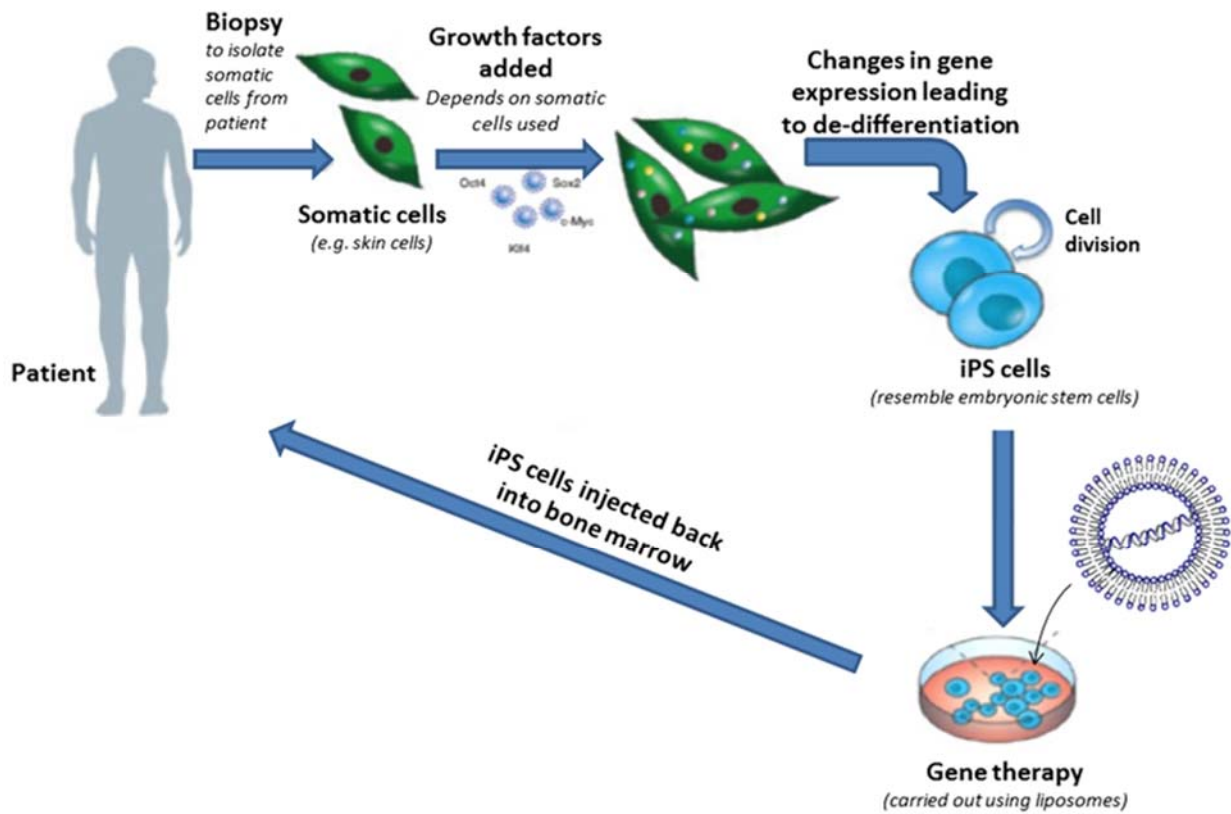
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.....[2]

(e) Suggest one reason why the number of cell colonies with minimal differentiation differs from the total number of cell colonies produced.

.....  
.....[1]

X-linked Severe Combined Immunodeficiency (SCID) is a rare congenital disorder characterised by improper development of immune cells which has been treated by gene therapy. The ability to generate iPS cells that have similar characteristics with embryonic stem cells has provided a promising alternative to the use of haematopoietic stem cells for gene therapy to treat X-linked SCID.

Fig 1.3 shows a possible process of using iPS cells for gene therapy.



**Fig. 1.3**

- (f) Besides difficulties with de-differentiating somatic cells to iPS cells, explain one other factor that could prevent this method of gene therapy for X-linked SCID from becoming an effective treatment.

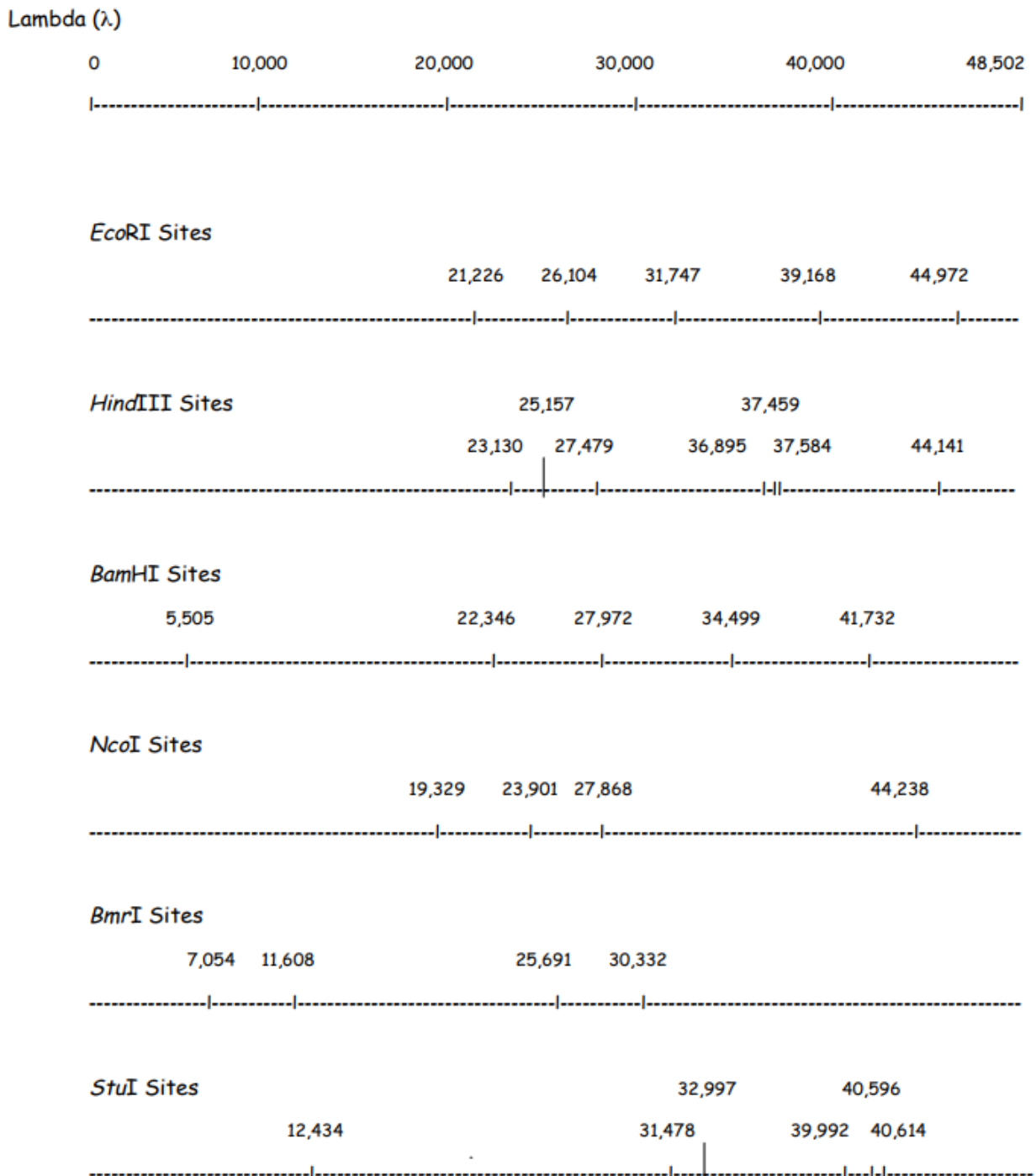
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- (g) Justify if you agree or disagree that use of iPS cells can address the ethical concerns of using embryonic stem cells for treatment of genetic diseases.

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 .....[2]

**[Total: 14]**

2. Conventional DNA ladders are traditional molecular weight standards used for sizing and approximate quantification of linear double-stranded DNA fragments in agarose and non-denaturing polyacrylamide gels for research purposes. The markers are composed of lambda phage DNA digested to completion with the appropriate restriction enzyme(s), purified and dissolved in storage buffer. The DNA fragments contain blunt or sticky ends depending on the restriction enzyme used for the marker's preparation. Fig. 2.1 shows the genome of Lambda DNA with the restriction sites corresponding to six different restriction enzymes.



**Fig. 2.1**

Table 2.1

| EcoRI | HindIII | NcoI | Bmrl |
|-------|---------|------|------|
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |
|       |         |      |      |

- (a) With reference to Fig. 2.1, fill in the columns of Table 2.1 with the respective DNA fragments generated from their corresponding restriction enzymes. List each fragment from the largest to the smallest. [2]
- (b) Explain two factors that would influence a researcher's decision in choosing a restriction enzyme for the restriction digest step of the gene cloning experiment.

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.....[2]

| Marker   | EcoRI<br>+ HindIII | NcoI<br>+ Bmrl |
|----------|--------------------|----------------|
| (50,000) |                    |                |
| (30,000) |                    |                |
| (20,000) |                    |                |
| (15,000) |                    |                |
| (10,000) |                    |                |
| (5,000)  |                    |                |
| (2,500)  |                    |                |
| (1,000)  |                    |                |

**Fig. 2.2**

- (c) Complete Fig. 2.2 below by drawing the DNA band patterns after gel electrophoresis. [2]



- (d) Explain how DNA bands can be visualized after gel electrophoresis.

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.....[2]

- (e) Suggest a method to improve the separation of DNA bands during gel electrophoresis.

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.....[1]

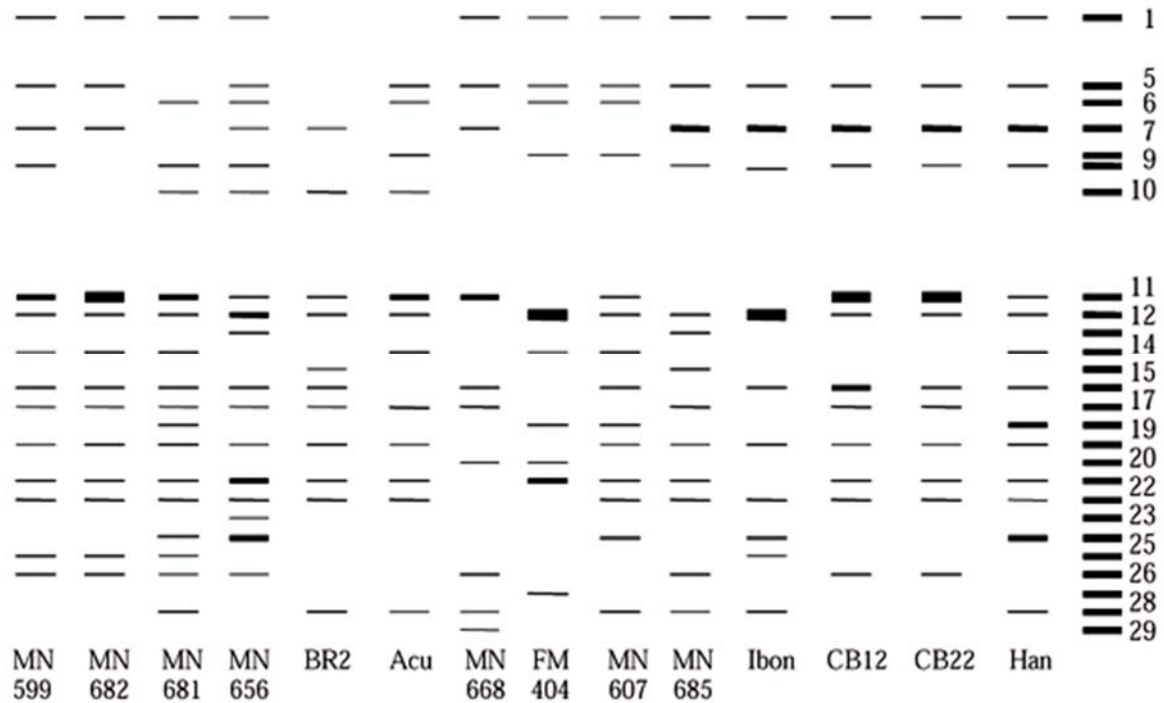
Besides being used for DNA fragment separation, gel electrophoresis can also be conducted to separate protein fragments for agricultural research purposes in order to study relationships between polypeptide band patterns and phenotypic traits.

Barley (*Hordeum vulgare*) is an important crop in southern Brazil where its production is used in the brewing industry. Hence, the malting quality of different barley plant cultivars must be continuously researched and improved upon. Cultivars are new plant species obtained via artificial selecting breeding processes.

Malt is germinated cereal grains that have been dried via malting. Malting grains develop the enzymes required for modifying the starch in the grains into various types of sugars such as maltose and glucose. Characteristics of importance for malting quality, which can differ considerably among barley cultivars, include grain size, grain protein concentration and nitrogen content in the seeds.

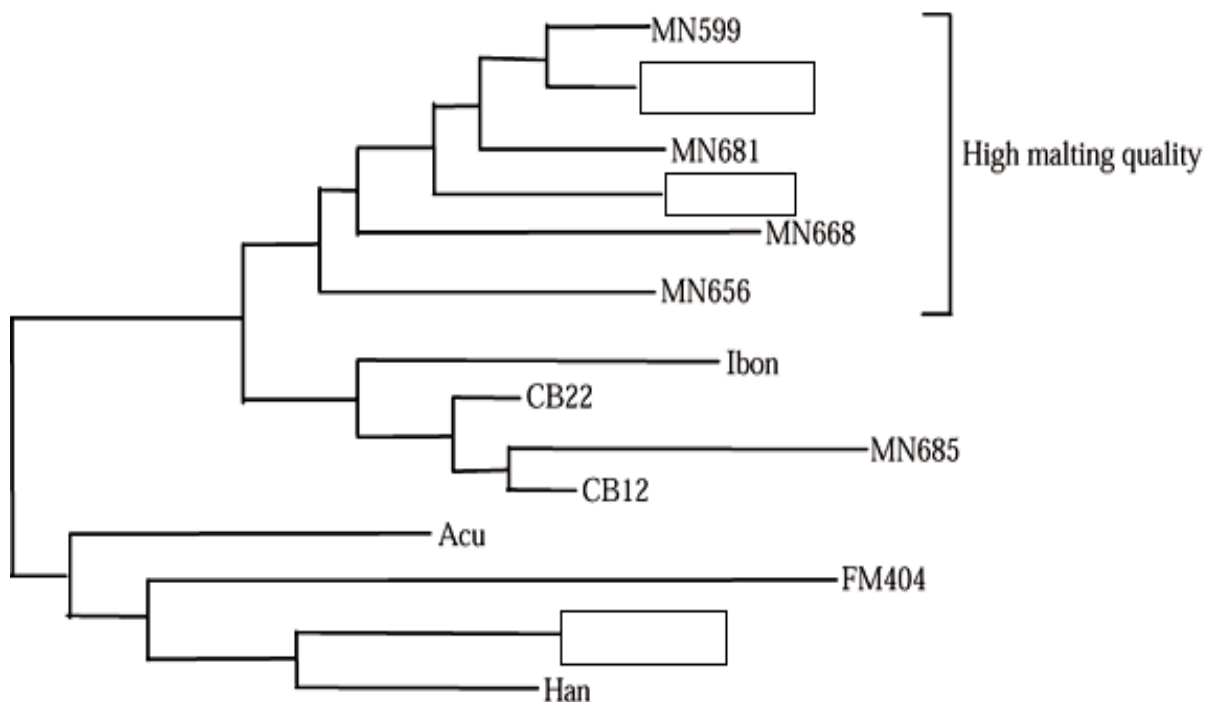
Recently, researchers are researching on how the quality of a particular storage protein named hordein could affect the malting quality of barley plants. Barley is highly polymorphic regarding the hordein polypeptide composition. Electrophoresis is used as a screening test to differentiate barley plant cultivars and to determine malting quality of each variety. By comparing the total hordein pattern from barley cultivars of different malting quality, researchers can investigate the relationship between malting quality and band patterns and to explore the feasibility of using hordein protein electrophoresis to assist in the selection of barley plant cultivars for malting.

Fig. 2.3 shows the hordein polypeptide band patterns of 14 different barley plant species. On the right of the gel, a polypeptide fragment ladder showing the band positions of 26 different hordein polypeptide sizes serves as a reference point for comparison.



**Fig. 2.3**

(f) With reference to Fig. 2.3, complete the phylogenetic tree shown in Fig. 2.4.



**Fig. 2.4**

[2]

Table 2.2 shows the correlation of each polypeptide band with the malting quality of barley varieties studied. The number of + / - in Table 2.2 indicates the strength of correlation between the polypeptide band and malting quality.

Table 2.2

| Band | Correlation | Band | Correlation |
|------|-------------|------|-------------|
| 1    | -           | 17   | -           |
| 5    | --          | 18   | -           |
| 6    | +           | 19   | ---         |
| 7    | -           | 20   | +           |
| 8    | -           | 21   | -           |
| 9    | -           | 22   | -           |
| 10   | +           | 23   | +           |
| 11   | +           | 24   | -           |
| 12   | --          | 25   | +           |
| 13   | +           | 26   | +           |
| 14   | -           | 27   | +           |
| 15   | +           | 28   | -           |
| 16   | +           | 29   | +           |

Table 2.3 shows the frequency of the hordein polypeptide bands (in percentage) in each barley variety studied.

Table 2.3

| Hordein<br>band | Varieties <sup>(1)</sup> |           |           |           |           |     |     |     |           |           |           |      |      |      |
|-----------------|--------------------------|-----------|-----------|-----------|-----------|-----|-----|-----|-----------|-----------|-----------|------|------|------|
|                 | MN<br>599                | MN<br>682 | MN<br>668 | MN<br>681 | MN<br>656 | BR2 | Acu | Han | FM<br>404 | MN<br>607 | MN<br>685 | Ibon | CB12 | CB22 |
| 1               | 20                       | 20        | 27        | 27        | 38        | 0   | 0   | 78  | 100       | 70        | 100       | 90   | 100  | 100  |
| 5               | 20                       | 10        | 18        | 0         | 8         | 0   | 91  | 71  | 100       | 70        | 100       | 90   | 100  | 90   |
| 6               | 0                        | 0         | 0         | 9         | 8         | 0   | 75  | 0   | 100       | 70        | 0         | 0    | 0    | 0    |
| 7               | 20                       | 20        | 27        | 0         | 31        | 13  | 0   | 78  | 0         | 0         | 100       | 90   | 100  | 100  |
| 8               | 0                        | 0         | 0         | 0         | 0         | 0   | 75  | 0   | 100       | 50        | 0         | 0    | 0    | 0    |
| 9               | 20                       | 0         | 0         | 27        | 8         | 0   | 0   | 64  | 0         | 0         | 100       | 70   | 100  | 90   |
| 10              | 0                        | 0         | 0         | 9         | 8         | 33  | 8   | 0   | 0         | 0         | 0         | 0    | 0    | 0    |
| 11              | 100                      | 100       | 100       | 100       | 54        | 100 | 100 | 100 | 0         | 100       | 0         | 0    | 100  | 100  |
| 12              | 100                      | 70        | 0         | 64        | 92        | 40  | 100 | 100 | 100       | 100       | 100       | 100  | 100  | 100  |
| 13              | 0                        | 0         | 0         | 0         | 46        | 0   | 0   | 0   | 0         | 0         | 100       | 0    | 0    | 0    |
| 14              | 30                       | 30        | 0         | 27        | 0         | 0   | 100 | 100 | 100       | 100       | 0         | 0    | 0    | 0    |
| 15              | 0                        | 0         | 0         | 0         | 0         | 60  | 0   | 0   | 0         | 0         | 100       | 0    | 0    | 0    |
| 16              | 100                      | 100       | 100       | 100       | 54        | 100 | 0   | 100 | 0         | 30        | 0         | 100  | 100  | 100  |
| 17              | 50                       | 60        | 100       | 73        | 23        | 60  | 91  | 0   | 0         | 0         | 100       | 0    | 100  | 100  |
| 18              | 0                        | 0         | 0         | 18        | 0         | 0   | 0   | 100 | 100       | 100       | 0         | 0    | 0    | 0    |
| 19              | 20                       | 20        | 0         | 36        | 54        | 66  | 100 | 100 | 0         | 90        | 100       | 100  | 100  | 100  |
| 20              | 0                        | 0         | 9         | 0         | 0         | 0   | 0   | 0   | 100       | 0         | 0         | 0    | 0    | 0    |
| 21              | 100                      | 100       | 0         | 100       | 46        | 100 | 42  | 100 | 100       | 100       | 100       | 0    | 100  | 100  |
| 22              | 100                      | 100       | 0         | 100       | 46        | 100 | 91  | 36  | 0         | 100       | 100       | 100  | 100  | 100  |
| 23              | 0                        | 0         | 0         | 0         | 23        | 0   | 0   | 0   | 0         | 0         | 0         | 0    | 0    | 0    |
| 24              | 0                        | 0         | 0         | 27        | 54        | 0   | 0   | 100 | 0         | 100       | 0         | 90   | 0    | 0    |
| 25              | 50                       | 30        | 0         | 45        | 0         | 0   | 0   | 0   | 0         | 0         | 0         | 20   | 0    | 0    |
| 26              | 40                       | 40        | 100       | 9         | 23        | 0   | 0   | 0   | 0         | 0         | 100       | 0    | 100  | 100  |
| 27              | 0                        | 0         | 0         | 0         | 0         | 0   | 0   | 0   | 100       | 0         | 0         | 0    | 0    | 0    |
| 28              | 0                        | 0         | 64        | 9         | 0         | 6   | 8   | 93  | 0         | 100       | 50        | 100  | 0    | 0    |
| 29              | 0                        | 0         | 27        | 0         | 0         | 0   | 0   | 0   | 0         | 0         | 0         | 0    | 0    | 0    |

- (g)** With reference to Fig. 2.4, Table 2.2 and Table 2.3, identify and explain which barley variant would have the best malting quality.

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**[Total: 15]**

3. Rice is the staple diet in many parts of the world. It lacks a number of important nutrients, including  $\beta$  carotene, from which vitamin A is synthesised. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease. Golden rice, which contains  $\beta$  carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

Fig. 3.1 shows an artificial DNA sequence used.



Key:

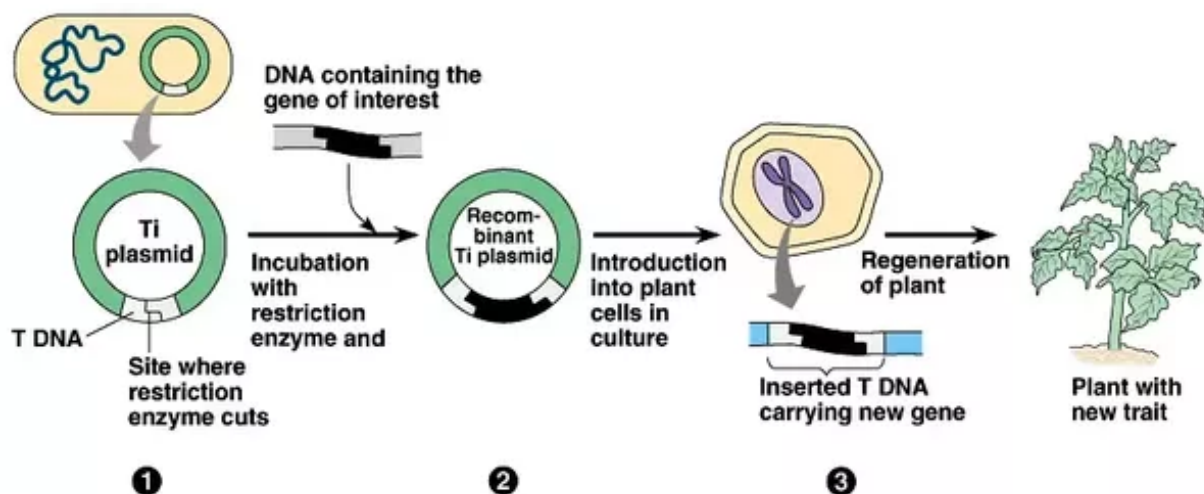
pro – promoter sequence for polymerase enzymes

ter – termination signal for polymerase enzymes

Hyg resist – Hygromycin B antibiotic resistance gene from a bacterium

**Fig. 3.1**

Fig. 3.2 shows the main events involved in obtaining a transgenic rice plant.



**Fig. 3.2**

- (a) Define the term 'transgenic'.

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 .....[1]

- (b)** Describe two unique and distinct regions found within the Ti plasmid.

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.....[2]

- (c)** With reference to Fig. 3.1 and Fig. 3.2,

- (i)** outline the processes involved from stage **2** to stage **3**.

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- (ii)** outline the processes involved from stage **3** to generate a full-grown transgenic plant.

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**[Total: 11]**

4. A pineapple plantation owner wants to find out the amount of ascorbic acid (vitamin C) in the pineapples cultivated in the plantation. He believes that his pineapples produce the most vitamin C compared to the standard pineapple breeds which typically contain 40.0 – 48.5 %.

The amount of ascorbic acid present in a sample can be determined using a bioassay method. At pH 7 and above, ascorbic acid reduces solutions of the dye dichlorophenol indophenol (DCPIP) from blue to colourless. For this experiment to work, the pH of the samples must be adjusted to pH 9. Ascorbic acid does not chemically change when neutralised by sodium hydroxide or when boiled.

Using this information and your own knowledge, design an experiment to determine the validity of the plantation owner's claim that the pineapples from his plantation contain higher concentrations of ascorbic acid.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you must use:

- 100 cm<sup>3</sup> of 5.0 % stock solution of ascorbic acid, adjusted to pH 7
- 100 cm<sup>3</sup> distilled water
- 100 cm<sup>3</sup> molten agar containing DCPIP
- Sterile petri dishes
- 1ml syringes
- Plastic straw to create wells in the agar plate
- Labels
- Stopwatch
- Forceps
- Normal laboratory glassware e.g. test tubes, beakers, graduated pipettes, droppers, glass rods etc
- 10 cm<sup>3</sup> pineapple juice, supplied by the plantation owner
- Sodium hydroxide
- pH meter

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of tables and graphs,
- use correct technical and scientific terms,
- include references to safety measures to minimize any risk associated with the proposed experiment.

**[Total: 12]**

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**Free-response question**

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**5.**

- (a)** Describe the use of restriction fragment length polymorphism analysis in creating a linkage map. **[6]**
- (b)** Outline the processes involved in PCR. **[6]**
- (c)** Discuss the goals and benefits of the Human Genome Project. **[8]**

**[Total: 20]**

**End of Paper**